Topic: Stack using Singly Linked List

Q. Explain how a Stack can be implemented using a Singly Linked List. Provide the logic for the Push and Pop operations.

> **Definition to Remember**

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dynam
stack.

Stack Implementation using Singly Linked List

To maintain O(1) time complexity for both Push and Pop operations, all insertions and deletions must occur at the **beginning (head)** of the linked list.

| Stack Operation | Linked List Equivalent |

| :--- | :

2. Push Operation (Insert at Beginning)

Adds a new item to the top of the stack.

Algorithm:

```text

### 3. Pop Operation (Delete from Beginning)

Removes and returns the item from the top of the stack.

**Algorithm:**

```text

### 4. Advantages of Linked List Implementation

- **Dynamic Size:** The stack can grow and shrink dynamically. It is limited only by system memory, unlike an array which has a fixed pre-defined size.

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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